Computational efficiency of multiresolution topology optimization for the MBB beam

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Abstract

This report assesses the computational efficiency of the multiresolution topology optimization (MTOP) approach of Nguyen et al. for the two-dimensional minimum-compliance MBB beam. The analysis is performed on a coarse finite element mesh while the design is represented on a finer density mesh. The convergence behavior, wall-clock time and optimized layout are compared with a classical density-based formulation at the same final density resolution. For the assignment specification of 600×200 density cells, the MTOP scheme reproduces the classical layout to within $0.06\,\%$ of fine-mesh compliance for sensitivity filtering and within $0.02\,\%$ for density filtering, while reducing the optimization time by an order of magnitude in both cases. The optimality-criteria update is then replaced by MMA and combined with sensitivity filtering, density filtering and Heaviside projection at three threshold values η. Heaviside projection produces near-binary designs whose fine-mesh compliance is about $17\,\%$ lower than the gray density-filtered design at the same volume fraction; the symmetric threshold $\eta = 0.5$ gives the lowest compliance of all the configurations tested.

1 Problem statement

The objective of the assignment is to assess whether the MTOP approach proposed by Nguyen et al. [2] reduces the computational cost of two-dimensional minimum-compliance topology optimization without altering the optimized layout. Two formulations are compared on the half MBB beam benchmark from the lecture material:

1. a classical density-based formulation in which the analysis mesh and the density mesh coincide, and
2. the MTOP formulation in which the analysis mesh is coarser than the density mesh.

The remaining studies of the assignment (density filtering, MMA, Heaviside projection) are listed in Section 6.1 and are reported in subsequent sections as the implementation progresses.

2 Structural model

2.1 Geometry, boundary conditions and loading

The structure is the half MBB beam used in the Chapter 9 lecture code [1]. A unit downward point load is applied to the upper-left corner. The left edge is restrained against horizontal motion (symmetry condition) and the lower-right corner is restrained against vertical motion (roller). Bilinear plane-stress quadrilateral elements with unit edge length, unit thickness, $E_0 = 1$ and Poisson's ratio $\nu = 0.3$ are used for the analysis.

2.2 Discretization

The classical baseline uses a finite element mesh of 600×200 elements. The MTOP model uses a coarse finite element (analysis) mesh of 120×40 elements with 5×5 density cells per analysis element, giving the same 600×200 density resolution as the classical model. The two discretizations are summarized in table 1.

Table 1: Discretization parameters of the classical and MTOP formulations.

Model	Analysis (FE) mesh	Density cells per element	Density mesh
Classical	600×200	1×1	600×200
MTOP	120×40	5×5	600×200

The number of free displacement degrees of freedom is $N_{\text{dof}}^{\text{cl}} \approx 2.41 \times 10^5$ for the classical mesh and $N_{\text{dof}}^{\text{mtop}} \approx 9.92 \times 10^3$ for the MTOP analysis mesh. The factor of about 24 in mesh size translates into a much larger reduction in the cost of solving $\mathbf{K}\mathbf{u} = \mathbf{f}$, which dominates the computation in this 2D setting.

2.3 Material interpolation

The Young's modulus depends on the physical density ρ through the SIMP law

$$E(\rho) = E_{\min} + \rho^p \,(E_0 - E_{\min}), \tag{1}$$

with penalization power $p = 3$, $E_0 = 1$ and $E_{\min} = 10^{-9}$. The volume fraction is fixed at $V^* = 0.5$.

3 Optimization problem

Let $\boldsymbol{x} \in [0,1]^N$ denote the vector of design variables and $\boldsymbol{\rho} = \mathcal{F}(\boldsymbol{x})$ the corresponding physical densities, where $\mathcal{F}$ is the filter operator. The minimum-compliance problem reads

$$\min_{\boldsymbol{x} \in [0,1]^N} \quad C(\boldsymbol{x}) = \mathbf{f}^T \mathbf{u}(\boldsymbol{\rho}), \tag{2}$$

$$\text{s.t.} \quad \mathbf{K}(\boldsymbol{\rho})\, \mathbf{u}(\boldsymbol{\rho}) = \mathbf{f}, \tag{3}$$

$$\frac{1}{N} \sum_{i=1}^{N} \rho_i \le V^*, \tag{4}$$

where N is the number of density cells. For sensitivity filtering one has $\rho_i = x_i$ and the filter is applied to the compliance sensitivity, while for density filtering $\rho_i = \sum_j H_{ij} x_j / \sum_j H_{ij}$ with cone weights $H_{ij} = \max(0, r_{\min} - \|\mathbf{x}_i - \mathbf{x}_j\|)$.

4 Solution technique

4.1 Classical formulation

The classical baseline is the standard SIMP density-based formulation [1]. The global stiffness matrix is assembled from the analytical bilinear Q4 element stiffness $\mathbf{K}_0$ as

$$\mathbf{K} = \sum_{e=1}^{N} \big[E_{\min} + \rho_e^p (E_0 - E_{\min})\big] \mathbf{K}_0^{(e)}. \tag{5}$$

The compliance sensitivity follows from the self-adjoint structure of the problem,

$$\frac{\partial C}{\partial \rho_e} = -p\,(E_0 - E_{\min})\,\rho_e^{p-1}\,\mathbf{u}_e^T \mathbf{K}_0 \mathbf{u}_e. \tag{6}$$

Sensitivity filtering with a cone kernel of radius $r_{\min} = 24$ density cells is applied to $\partial C/\partial \rho_e$ following the convention used in the lecture code, see Section 4.3. The volume constraint is enforced by the optimality criteria (OC) update with bisection on the Lagrange multiplier; the move limit is $m = 0.2$ and the convergence tolerance on the maximum design change is $\varepsilon = 0.01$.

4.2 Multiresolution formulation

In the MTOP formulation each analysis element e contains $N_\mathrm{c} = 25$ density cells. Following Nguyen et al. [2], the element stiffness matrix is integrated by midpoint quadrature with one integration point at the centre of every density cell:

$$\mathbf{K}_e = \sum_{k=1}^{N_\mathrm{c}} \left[E_{\min} + \rho_{ek}^p (E_0 - E_{\min})\right] \mathbf{I}_k, \tag{7}$$

where ρ_{ek} denotes the density of cell k inside element e and

$$\mathbf{I}_k = A_k\, \mathbf{B}^T(\boldsymbol{\xi}_k)\, \mathbf{D}_0\, \mathbf{B}(\boldsymbol{\xi}_k). \tag{8}$$

$\mathbf{B}(\boldsymbol{\xi}_k)$ is the strain–displacement matrix of the bilinear Q4 shape functions evaluated at the cell centre $\boldsymbol{\xi}_k$, $\mathbf{D}_0$ is the plane-stress constitutive matrix at $E_0 = 1$, and $A_k = 1/N_\mathrm{c}$ is the area of one density cell. The 25 templates $\mathbf{I}_k$ are precomputed once per run; their sum approximates the analytical Q4 stiffness $\mathbf{K}_0$ to the order of the midpoint rule.

The compliance sensitivity with respect to the density of cell k in element e follows directly from the chain rule applied to eq. (7),

$$\frac{\partial C}{\partial \rho_{ek}} = -p\,(E_0 - E_{\min})\,\rho_{ek}^{p-1}\,\mathbf{u}_e^T\, \mathbf{I}_k\, \mathbf{u}_e. \tag{9}$$

This expression replaces the element-wise strain-energy density of eq. (6) by a per-cell quantity that depends on the local position of the cell within the analysis element. The volume sensitivity is uniform: $\partial V/\partial \rho_{ek} = 1/N$.

4.3 Filtering

The cone kernel $H_{ij} = \max(0, r_{\min} - \|\mathbf{x}_i - \mathbf{x}_j\|)$ acts on the density grid and is identical for the classical and MTOP runs. With the lecture convention used in `ex1a.m`, the sensitivity filter applied to eq. (6) or eq. (9) reads

$$\widetilde{\frac{\partial C}{\partial x_i}} = \frac{1}{\max(\gamma, x_i)}\, \frac{\sum_j H_{ij}\, x_j\, (\partial C/\partial x_j)}{\sum_j H_{ij}}, \qquad \gamma = 10^{-3}. \tag{10}$$

A filter radius of $r_{\min} = 24$ density cells is used both for the classical mesh and for the MTOP density mesh, so that the two formulations enforce the same minimum length scale.

4.4 Optimality criteria update

The OC update is identical for both formulations and uses the standard form

$$x_i^{\mathrm{new}} = \max\!\left[0,\, \max(x_i - m,\, \min(1,\, \min(x_i + m,\, x_i\sqrt{B_i/\lambda})))\right], \qquad B_i = -\frac{\widetilde{\partial C/\partial x_i}}{\partial V/\partial x_i}, \tag{11}$$

with $m = 0.2$ and λ adjusted by bisection until the volume constraint is active.

4.5 MMA and Heaviside projection

In step 4 the OC update is replaced by the method of moving asymptotes (MMA) of Svanberg, which approximates the objective and constraint by separable convex functions of moving-asymptote-shifted variables and solves the resulting subproblem by an interior-point method. The implementation `mma.m` that ships with the lecture toolbox is used unmodified. The objective passed to MMA is $f_0 = C/100$ and the volume constraint is $f_1 = v/V^* - 1 \leq 0$; the corresponding gradients are $\partial C/\partial x/100$ and $\partial v/\partial x/V^*$. The design variables remain in $[0, 1]$ as before.

For the Heaviside-projection variant, density filtering is followed by a smooth projection of Wang, Lazarov and Sigmund towards $\{0, 1\}$,

$$\rho_i = \frac{\tanh(\beta\eta) + \tanh\big(\beta(\widetilde{\rho}_i - \eta)\big)}{\tanh(\beta\eta) + \tanh\big(\beta(1 - \eta)\big)}, \tag{12}$$

where $\widetilde{\rho}_i = \mathcal{F}(\boldsymbol{x})_i$ is the density-filtered field, $\eta \in (0, 1)$ is the projection threshold and $\beta \geq 0$ controls the projection sharpness. The compliance and volume sensitivities pick up the additional factor

$$\frac{\partial \rho_i}{\partial \widetilde{\rho}_i} = \frac{\beta\,\mathrm{sech}^2\big(\beta(\widetilde{\rho}_i - \eta)\big)}{\tanh(\beta\eta) + \tanh(\beta(1 - \eta))}, \tag{13}$$

which multiplies $\partial C/\partial \rho_i$ before the linear filter chain rule is applied to obtain $\partial C/\partial x_k$. Continuation on β is used to avoid local minima at high projection sharpness: β starts at 1 and is doubled every 50 iterations until reaching $\beta_{\max} = 16$.

5 Sensitivity verification

Both formulations are verified before the production runs by comparing the analytical sensitivities with central finite differences. The verification is performed on a reduced MBB instance (12×4 analysis elements, 5×5 density cells per element) with random densities perturbed about the volume fraction. The compliance and the volume are evaluated by the same routines used inside the optimization loop. Twenty design variables are sampled at random and their analytical and finite-difference compliance sensitivities are compared with a central step of $h = 10^{-6}$. With this step the truncation and round-off errors of the central difference scheme are both controlled, so an analytical sensitivity that is exact to working precision should agree with the finite-difference reference well below the percent level. The verification routine `verify_sensitivities.m` stores the sample values, the maximum absolute and relative errors and the test settings in `sensitivity_verification.mat` and is executed automatically at the start of `run_assignment.m`. For the present runs the maximum relative error in the compliance sensitivity is 5.7×10^{-7} for the classical formulation and 1.5×10^{-5} for the MTOP formulation; the volume sensitivity is recovered to machine precision in both cases. The slightly larger MTOP error reflects the accumulation of $N_\mathrm{c} = 25$ midpoint contributions per element, but it is still many orders of magnitude smaller than any source of error that could affect the optimization trajectory.

6 Numerical experiments

6.1 Experiment matrix

6.2 Quantities recorded per iteration

For each run the iteration counter, the elapsed wall-clock time, the compliance, the volume fraction and the maximum design change are stored. All runs use a single thread on the same

Table 2: Planned numerical experiments. Steps marked with a $\star$ are reported in this version of the document.

Case	Approach	Optimizer	Filter / projection	Status
1	Classical	OC	Sensitivity filter	$\star$
2	MTOP	OC	Sensitivity filter	$\star$
3	Classical	OC	Density filter	$\star$
4	MTOP	OC	Density filter	$\star$
5	MTOP	MMA	Sensitivity filter	$\star$
6	MTOP	MMA	Density filter	$\star$
7	MTOP	MMA	Heaviside projection, $\eta \in \{0.3, 0.5, 0.7\}$	$\star$

machine and the same MATLAB release; this is the standard practice in MTOP studies, where comparing speedup factors across hardware is not meaningful.

7 Results

7.1 Classical OC baseline

The classical OC baseline of step 1 converges in 94 iterations. The final compliance is $C_{\text{cl}} = 224.596$, the volume fraction is 0.50007 and the maximum design change at convergence is 9.95×10^{-3}. The wall-clock time on the reference machine is 241.99 s. The optimized layout (fig. 1) shows the typical truss-like topology of the half MBB beam: a top chord in compression near the load line, a bottom chord in tension near the support line, and a system of inclined web members carrying the shear toward the support. The convergence histories as functions of the iteration number and of the wall-clock time are shown in figs. 2 and 3.

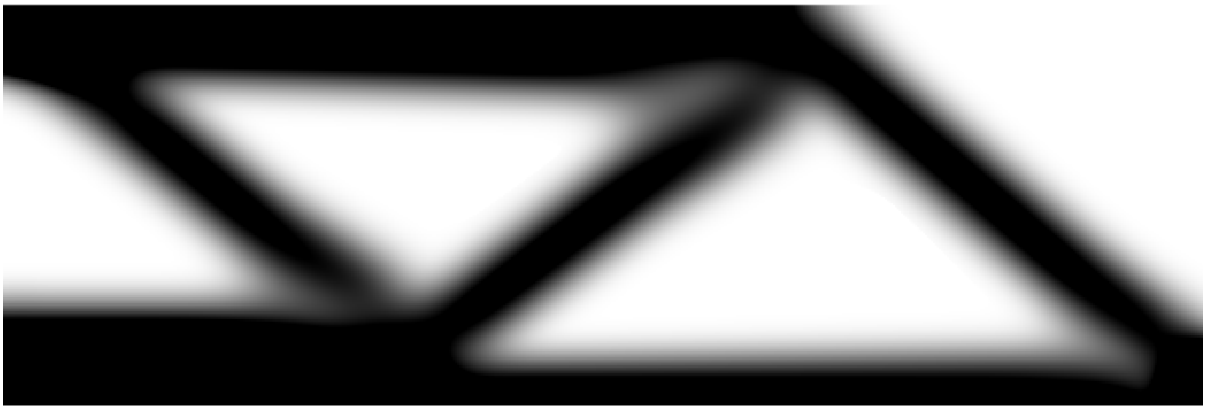

Figure 1: Optimized classical OC design with sensitivity filtering on a 600×200 mesh, $V^* = 0.5$, $p = 3$ and $r_{\min} = 24$ density cells. Black represents solid material, white represents void.

7.2 MTOP OC with sensitivity filtering

The MTOP OC run reaches the same convergence tolerance in 94 iterations, identical to the classical baseline. The MTOP compliance evaluated on the coarse 120×40 analysis mesh with the multiresolution stiffness matrix (7) is $C_{\text{mtop}}^{\text{coarse}} = 219.352$. To ensure a comparable measure of mechanical performance, the final MTOP density field is also assembled and analyzed on the same 600×200 classical finite element model: this gives a fine-mesh compliance of $C_{\text{mtop}}^{\text{fine}} = 224.731$, only 0.0601 % above the classical compliance C_{cl}. The volume fraction at

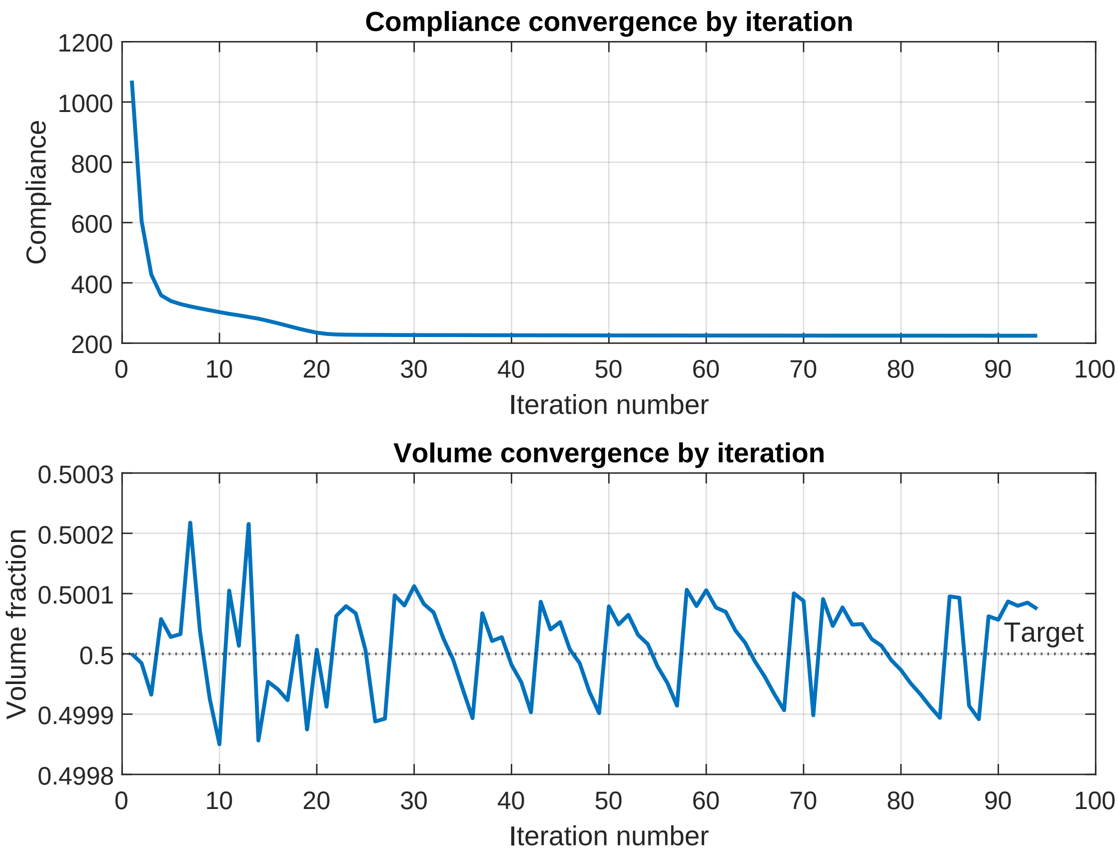

Figure 2: Classical OC convergence history versus iteration number: compliance (top) and volume fraction (bottom).

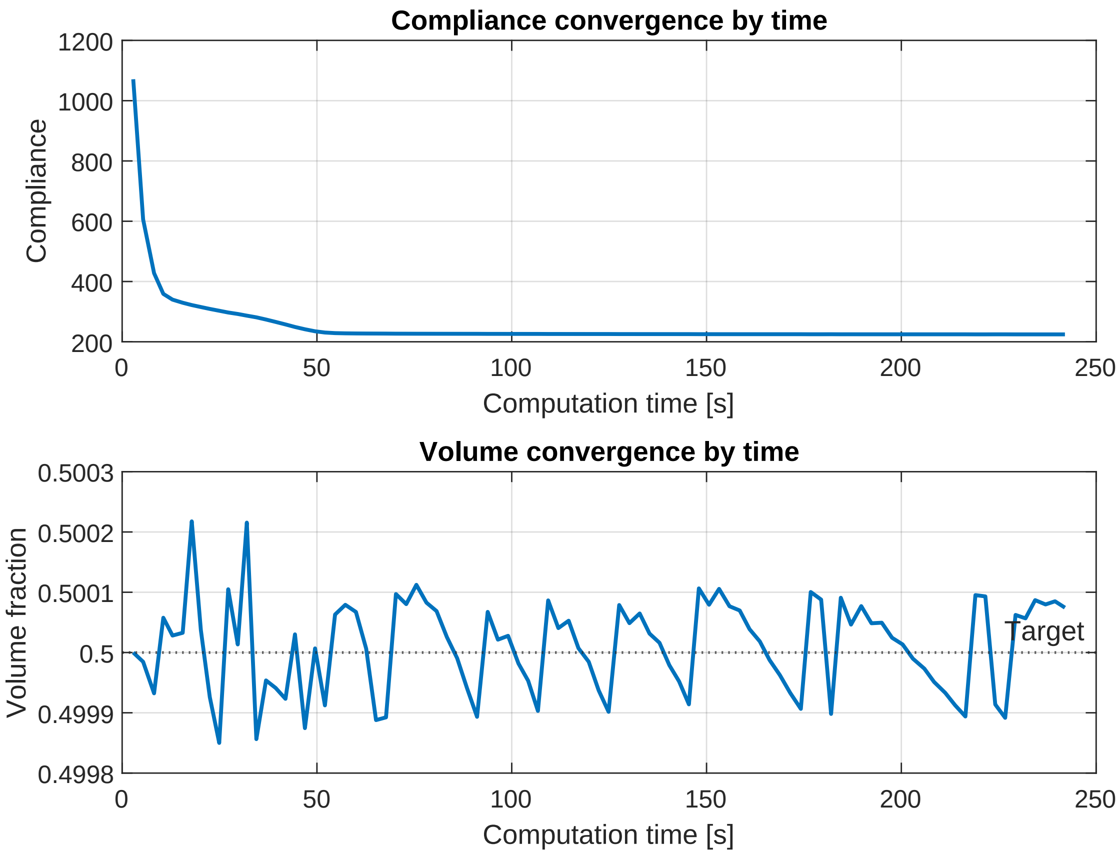

Figure 3: Classical OC convergence history versus elapsed wall-clock time: compliance (top) and volume fraction (bottom). The same data are used in fig. 7 for the runtime comparison with MTOP.

convergence is 0.500001 and the wall-clock time is 13.43 s, which corresponds to a speedup of 18.01 relative to the classical baseline.

Figure 4: Optimized MTOP OC design with a 120×40 analysis mesh and 5×5 density cells per analysis element.

7.3 Classical OC with density filtering

The third experiment replaces the sensitivity filter by a density filter while keeping all other parameters of step 1 unchanged. The compliance drops to a stable plateau within roughly fifty iterations, but the maximum design change does not fall below the convergence threshold $\varepsilon = 0.01$ within the budget of 300 iterations: it stagnates between 0.04 and 0.07 and the optimizer cycles between two near-equivalent layouts. The compliance at the iteration cap is $C_{\mathrm{cl}}^{\mathrm{dens}} = 240.976$, the volume fraction is 0.500 and the wall-clock time is 867.92 s. The corresponding design (fig. 8) shows the same load-path topology as the sensitivity-filtered design but with notably wider grey transitions across each member boundary, which is the hallmark of density filtering: the physical density is the smoothed version of the design variable, so the boundary cannot be sharper than one filter radius.

7.4 MTOP OC with density filtering

The MTOP run with density filtering also reaches the iteration cap of 300 without satisfying the design-change tolerance, with the same oscillation pattern observed for the classical density-filtered run. The native (multiresolution) compliance at the iteration cap is $C_{\mathrm{mtop}}^{\mathrm{coarse,dens}} = 235.621$ and the corresponding fine-mesh compliance is $C_{\mathrm{mtop}}^{\mathrm{fine,dens}} = 241.028$, only 0.0215 % above the classical density-filtered compliance. The volume fraction at the iteration cap is 0.500 and the wall-clock time is 137.25 s, giving a speedup of 6.32 over the classical density-filtered baseline.

7.5 MTOP MMA: sensitivity and density filtering

Step 4 keeps the multiresolution analysis mesh of step 2 and the same filter radius, but replaces the OC update by MMA. The first two MMA experiments use sensitivity filtering and density filtering, which are directly comparable with the corresponding OC runs of steps 2 and 3.

MMA with sensitivity filtering. The run satisfies the change tolerance $\varepsilon = 0.01$ in 188 iterations and gives a final compliance of $C_{\mathrm{mma,sens}}^{\mathrm{coarse}} = 221.085$ on the coarse analysis mesh and $C_{\mathrm{mma,sens}}^{\mathrm{fine}} = 226.452$ on the fine 600×200 analysis mesh. The volume fraction at convergence is 0.499999 and the wall-clock time is 129.51 s. The MTOP MMA design with sensitivity filtering is therefore very close to its OC counterpart of step 2, but reached after roughly twice as many iterations and with a slightly higher fine-mesh compliance (+0.77 % vs. $C_{\mathrm{mtop}}^{\mathrm{fine}} = 224.731$).

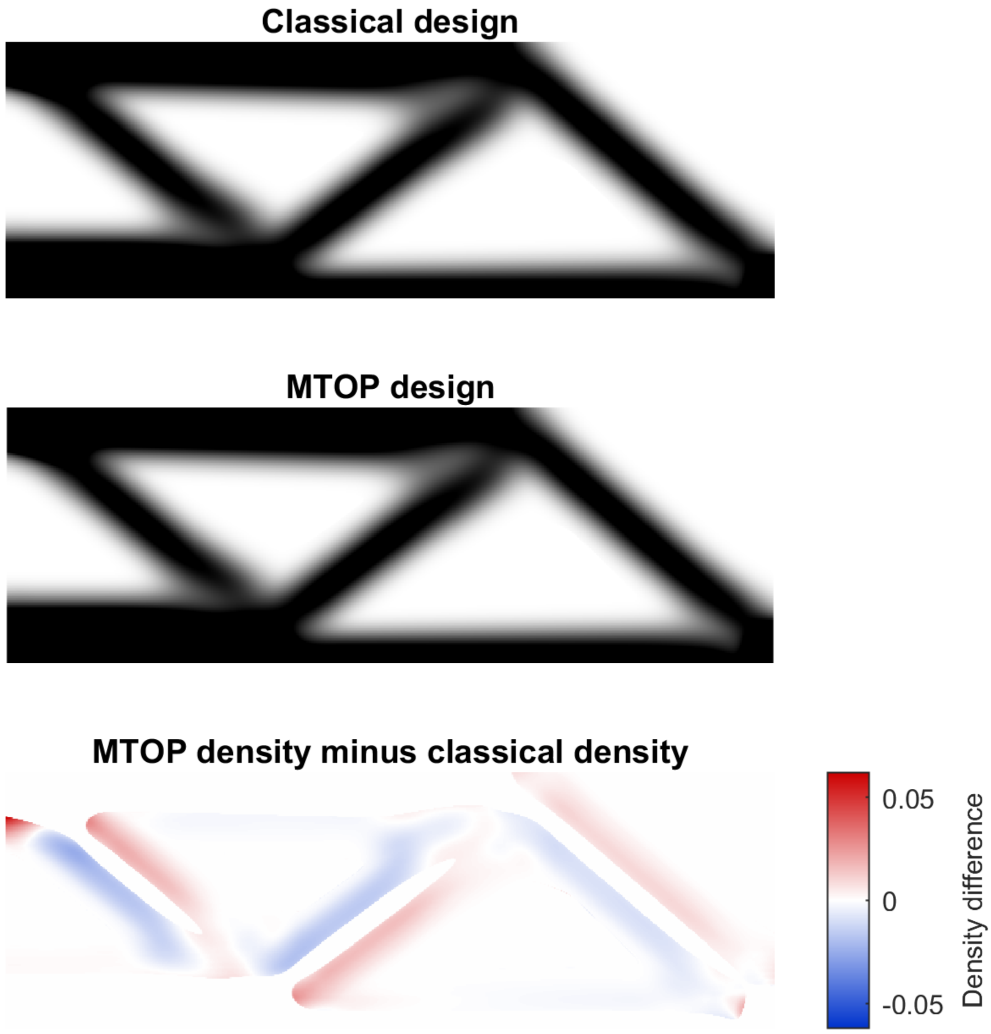

Figure 5: Comparison of the classical and MTOP OC designs (top, middle) and signed density difference $\rho_{\mathrm{mtop}} - \rho_{\mathrm{cl}}$ (bottom). The differences are concentrated on the boundaries of the structural members, where the per-cell midpoint integration of the multiresolution stiffness matrix differs most from the analytical Q4 integration of the classical formulation. The maximum absolute density difference is 0.0623, the RMS difference is 0.0051 and the mean absolute density difference is 0.0024.

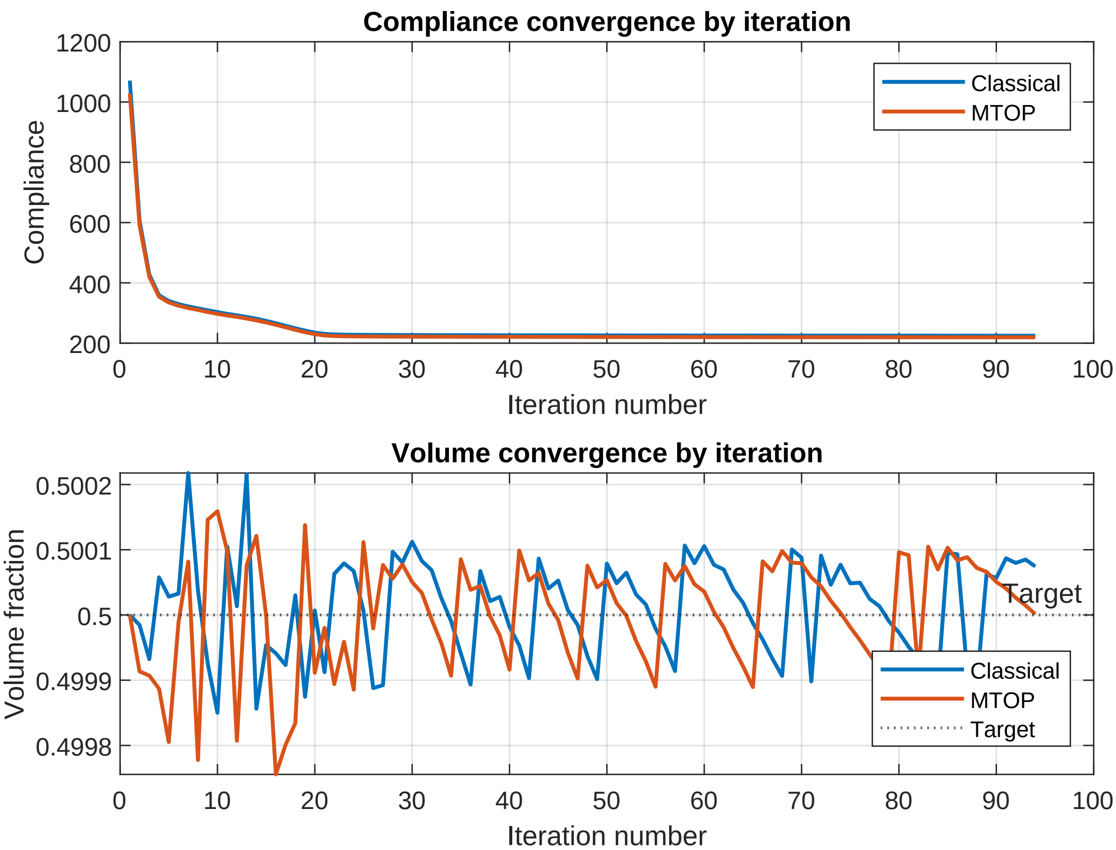

Figure 6: Classical and MTOP OC convergence histories versus iteration number. The two histories are nearly indistinguishable.

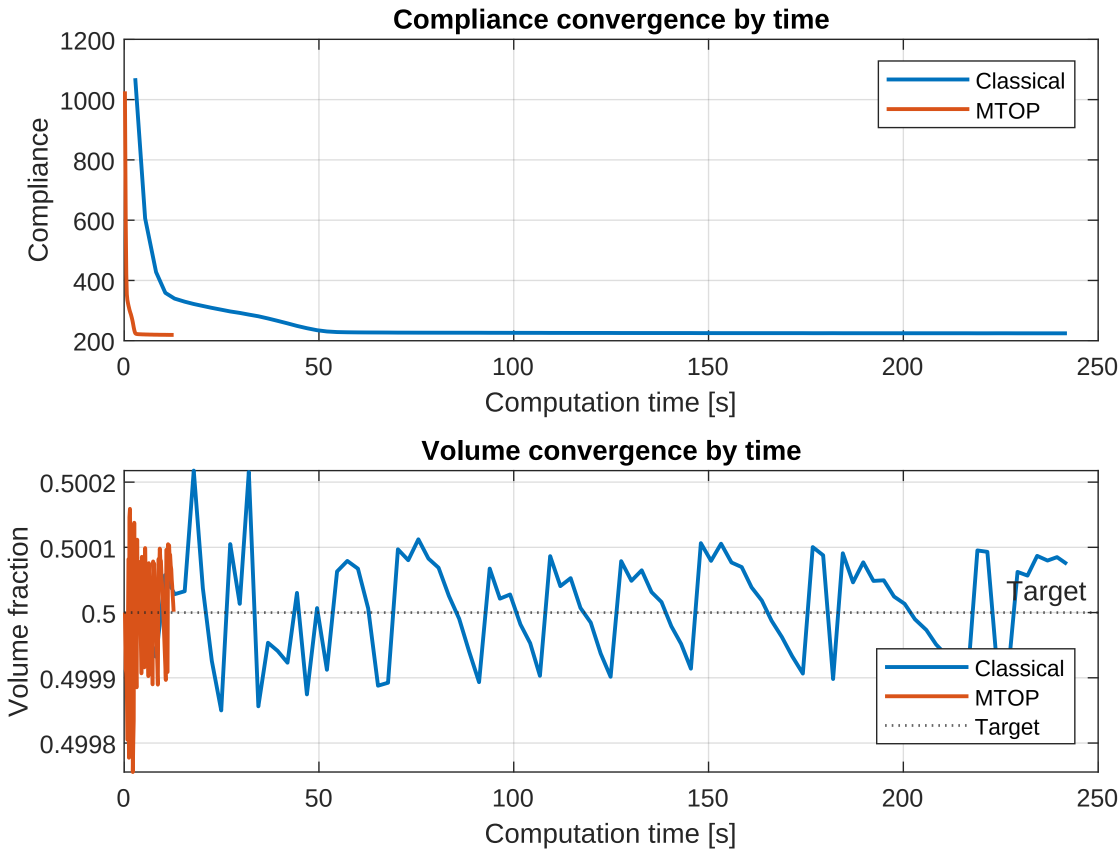

Figure 7: Classical and MTOP OC convergence histories versus wall-clock time. The MTOP curve is compressed against the left axis because each iteration solves a much smaller equilibrium problem.

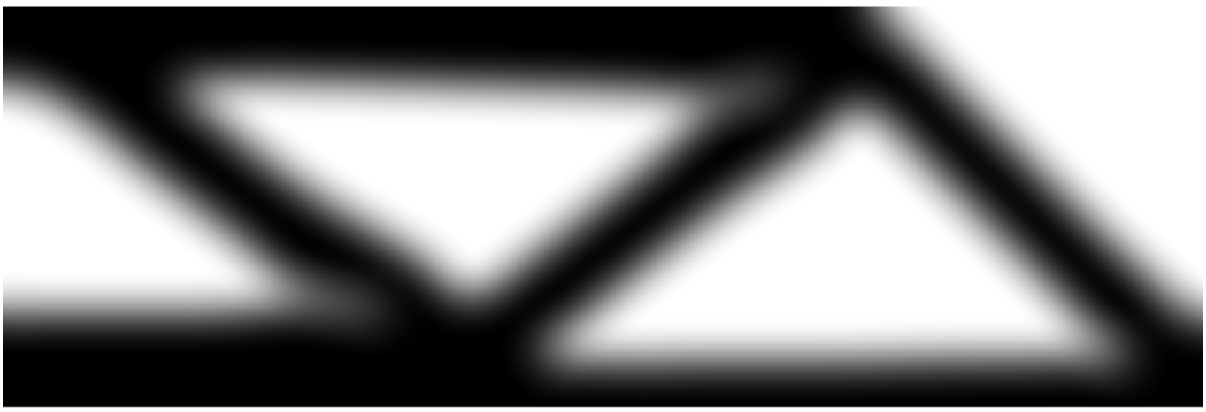

Figure 8: Optimized classical OC design with density filtering on a 600×200 mesh, $V^* = 0.5$, $p = 3$ and $r_{\min} = 24$ density cells. The grey halo around each member width corresponds to the filter radius and reflects the minimum length scale enforced by the density filter.

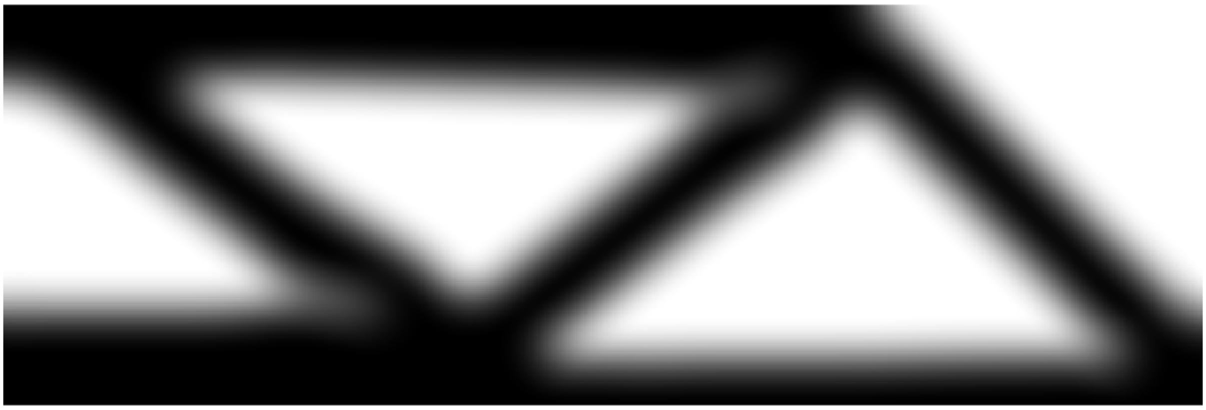

Figure 9: Optimized MTOP OC design with density filtering, on a 120×40 analysis mesh and 5×5 density cells per analysis element.

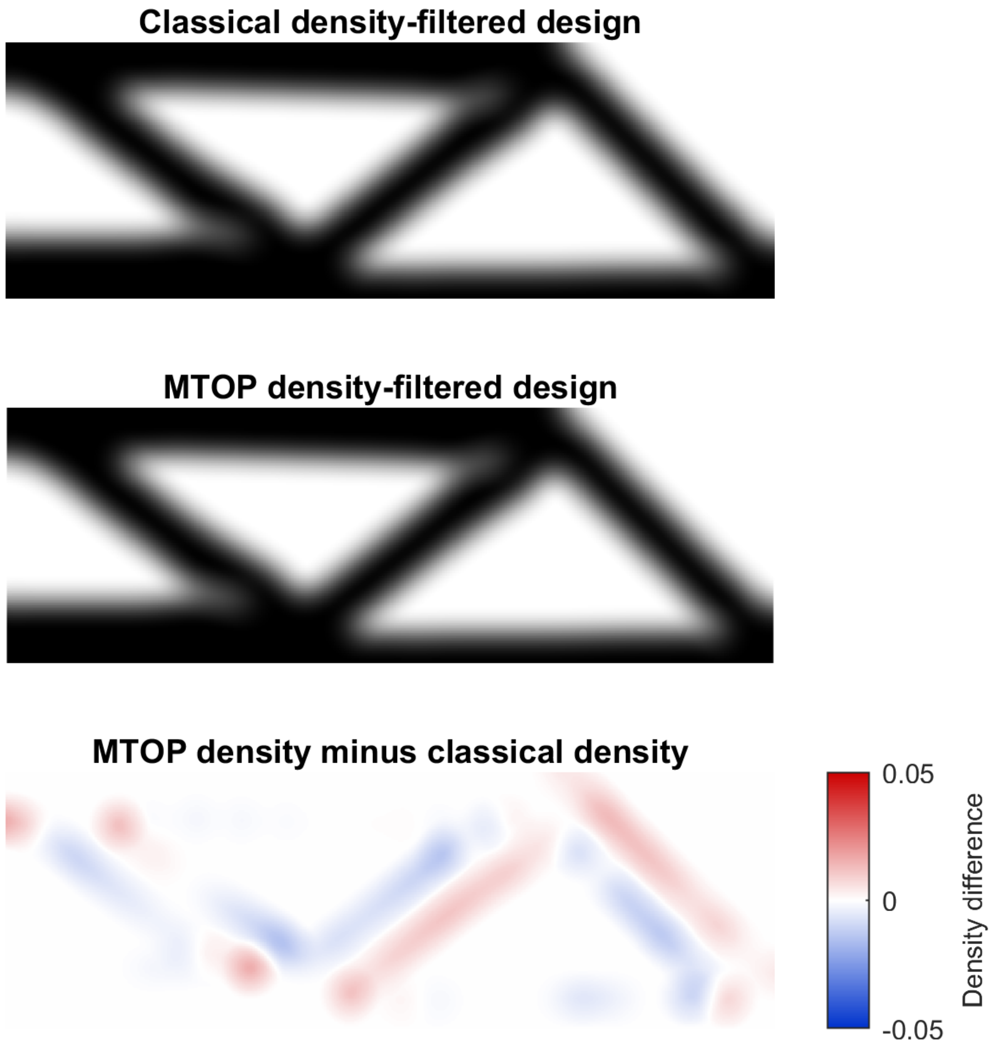

Figure 10: Comparison of the classical and MTOP OC designs with density filtering (top, middle) and signed density difference $\rho_{\mathrm{mtop}} - \rho_{\mathrm{cl}}$ (bottom). The maximum absolute density difference is 0.0159, the RMS difference is 0.0037 and the mean absolute density difference is 0.0021. The differences are markedly smaller than for sensitivity filtering (fig. 5) because the density filter pre-smooths the design variables before the SIMP analysis, so the choice of integration scheme inside the analysis element has less effect on the resulting layout.

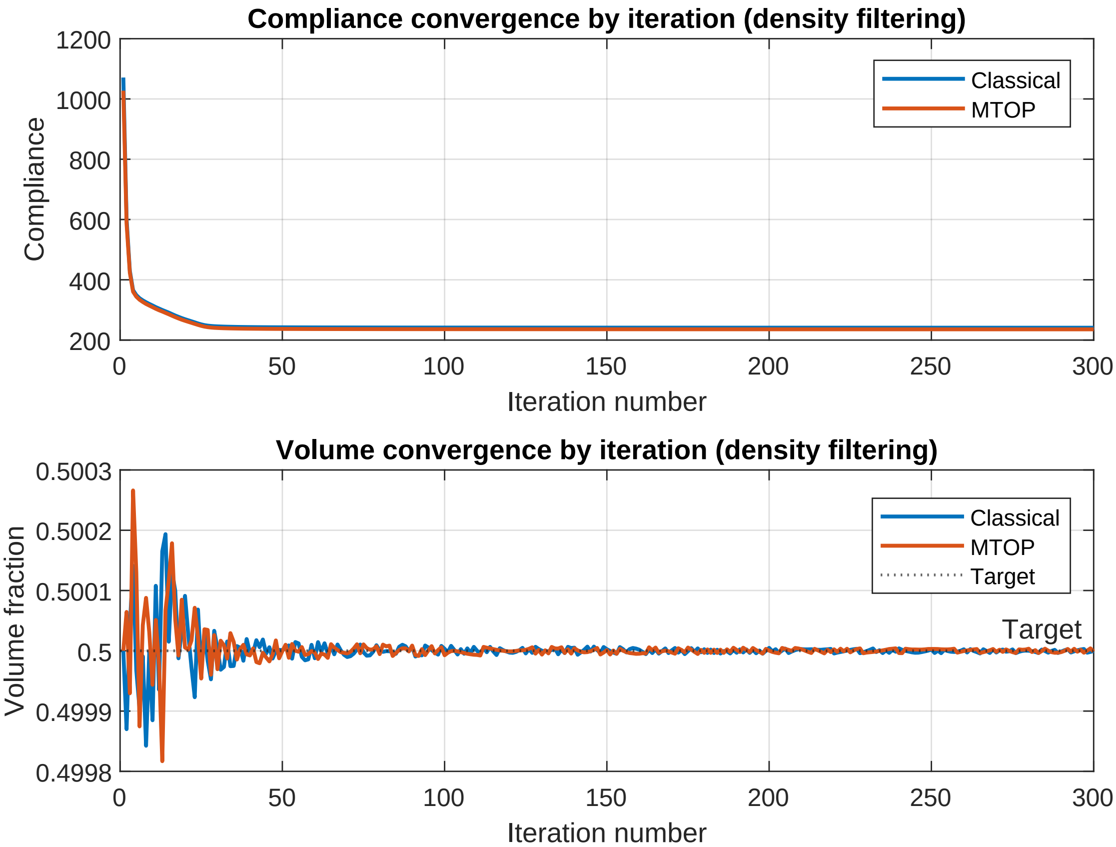

Figure 11: Classical and MTOP OC convergence histories versus iteration number, for density filtering. The compliance is well-stabilized after about fifty iterations, but the maximum design change cannot reach the $\varepsilon = 0.01$ tolerance within the iteration budget for either formulation.

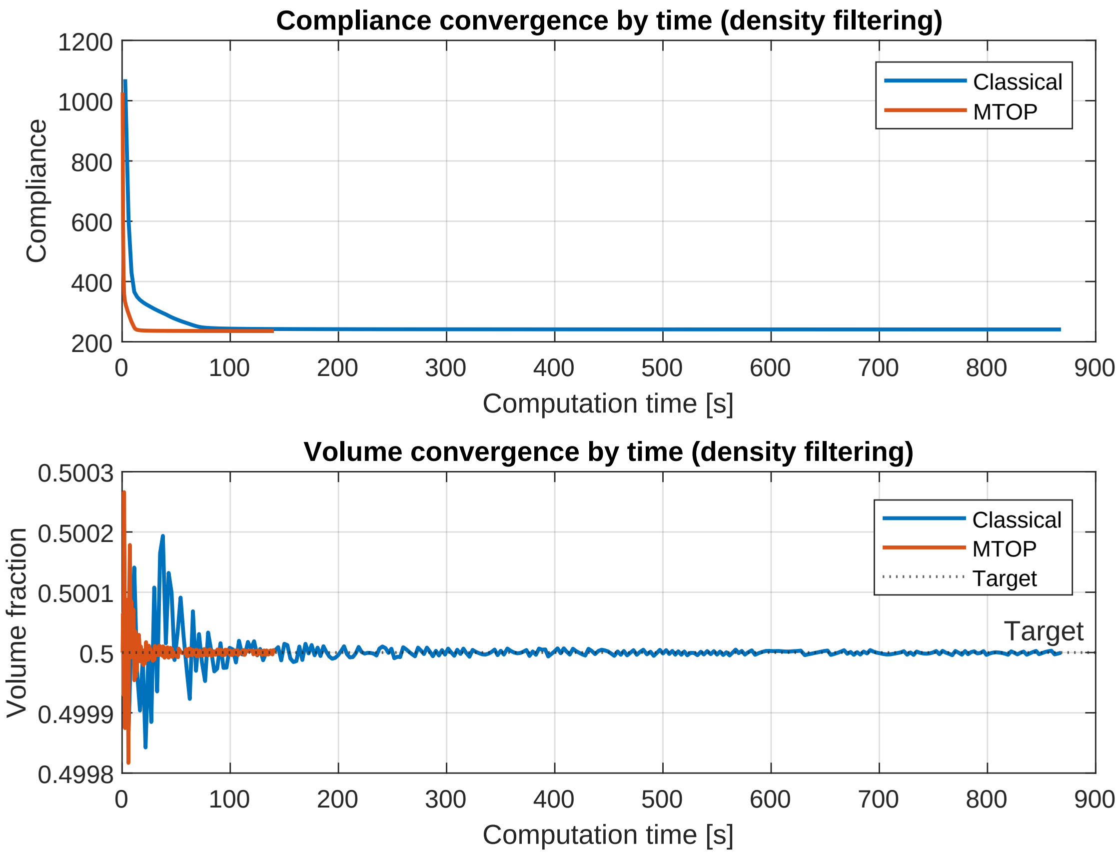

Figure 12: Classical and MTOP OC convergence histories versus wall-clock time, for density filtering. Both runs use 300 iterations because the change tolerance is not met; the speedup ratio is therefore the same as the per-iteration speedup, which is 6.32 for this run.

MMA with density filtering. The run reaches the iteration cap of 300 without satisfying the change tolerance: the maximum design change at termination is 0.055, while the compliance is well-stabilized at $C^{\text{coarse}}_{\text{mma,dens}} = 234.860$ on the coarse analysis mesh and $C^{\text{fine}}_{\text{mma,dens}} = 240.253$ on the fine analysis mesh. The volume fraction at termination is 0.499999 and the wall-clock time is 177.64 s. The fine-mesh compliance lies 0.31 % below the OC density-filtered value of 240.976 from step 3, so MMA does produce a marginally better design, but the change-tolerance issue of OC density is not resolved by switching the optimizer alone.

Figure 13: Optimized MTOP MMA design with sensitivity filtering.

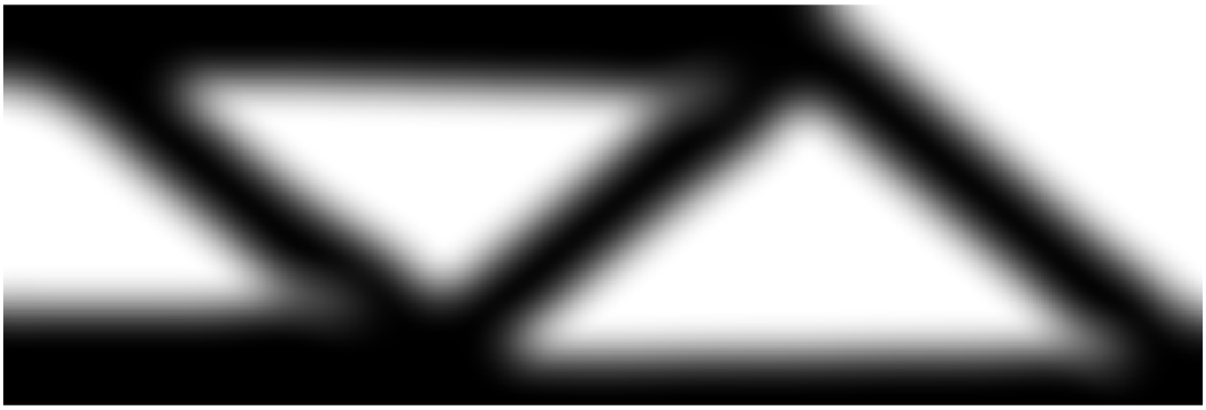

Figure 14: Optimized MTOP MMA design with density filtering.

7.6 MTOP MMA with Heaviside projection

Heaviside projection is applied on top of density filtering with three threshold values $\eta \in \{0.3, 0.5, 0.7\}$. For each η, the continuation strategy described in Section 4.5 is used, doubling β from 1 to 16 every 50 iterations. Final compliances, iteration counts and wall-clock times are reported in table 3; the optimized designs are shown in fig. 15.

7.7 Runtime comparison

8 Discussion

8.1 Convergence behavior

The classical OC iteration history shows the characteristic behavior of penalized density-based topology optimization. The compliance drops by almost a factor of five during the first ten

$\eta = 0.3$ (solid-biased threshold) $\eta = 0.5$ (symmetric threshold) $\eta = 0.7$ (void-biased threshold)

Figure 15: Optimized MTOP MMA designs with Heaviside projection for the three threshold values η used in step 4.

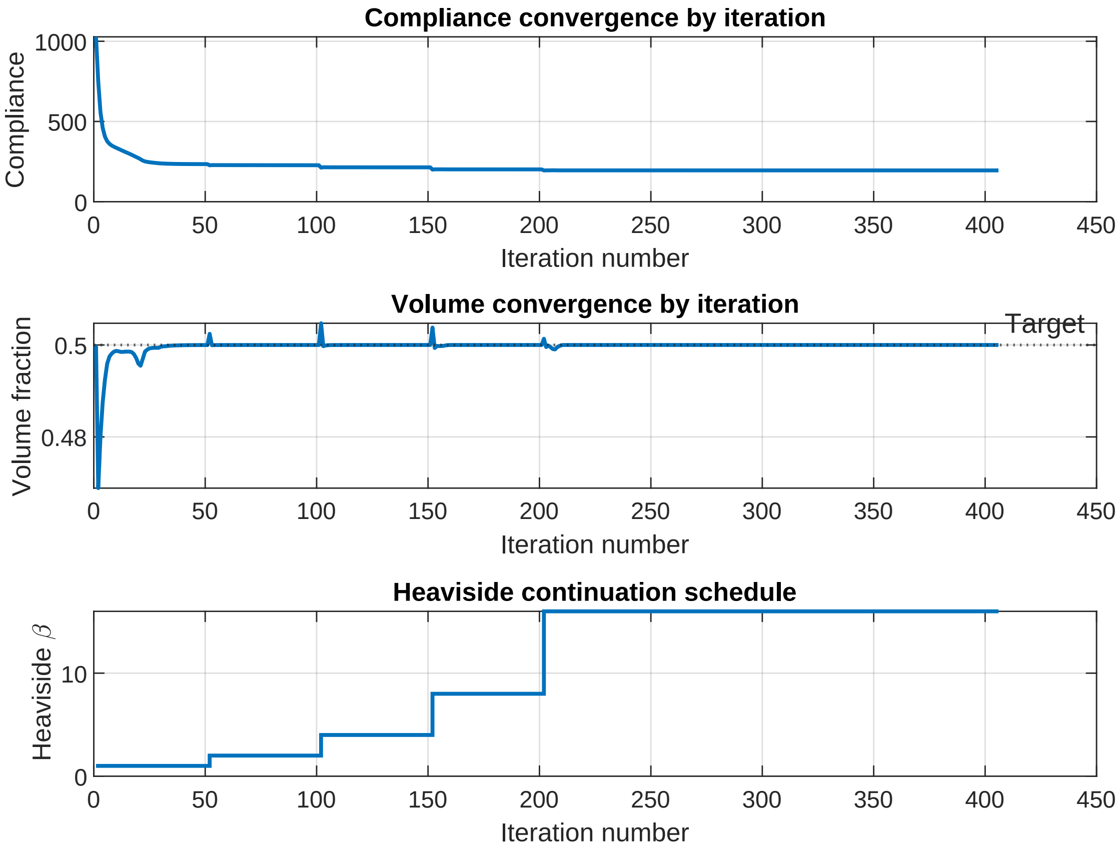

Figure 16: Convergence history for the symmetric Heaviside run ($\eta = 0.5$): compliance, volume fraction and the continuation parameter β versus iteration number. Each β doubling corresponds to a small jump in the compliance because the projection becomes sharper. The final design is reached at $\beta = \beta_{\max} = 16$.

Table 3: Wall-clock time and final objective values. “Native compliance” refers to the compliance computed inside each formulation (classical fine-mesh compliance for the classical case, multiresolution coarse-mesh compliance for the MTOP case). “Fine compliance” is the compliance of the optimized density field re-evaluated on the 600×200 classical analysis model. The dagger (†) marks runs that hit the iteration cap of 300 without satisfying the design-change tolerance $\varepsilon = 0.01$. Empty rows correspond to experiments that are not yet implemented.

Case	Iter.	Time [s]	Native C	Fine C	Volume
Classical OC sensitivity	94	241.99	224.596	224.596	0.500074
MTOP OC sensitivity	94	13.43	219.352	224.731	0.500001
Classical OC density	300†	867.92	240.976	240.976	0.500000
MTOP OC density	300†	137.25	235.621	241.028	0.499999
MTOP MMA sensitivity	188	129.51	221.085	226.452	0.499999
MTOP MMA density	300†	177.64	234.860	240.253	0.499999
MTOP MMA Heaviside, $\eta = 0.3$	364	217.68	195.633	200.129	0.499999
MTOP MMA Heaviside, $\eta = 0.5$	406	234.77	195.028	199.472	0.499999
MTOP MMA Heaviside, $\eta = 0.7$	500†	268.08	196.142	200.699	0.500000

iterations, when the uniform initial density redistributes towards a load-carrying skeleton, and then approaches a plateau slowly determined by SIMP penalization and sensitivity filtering. The volume fraction is held within $\pm 2 \times 10^{-4}$ of the prescribed target throughout the optimization by the OC bisection. The MTOP run reaches convergence in the same number of iterations as the classical baseline (94), which is consistent with the observation in [2] that MTOP affects the cost of each iteration but not the optimization trajectory itself.

8.2 Topological similarity of the two designs

The MTOP and classical layouts are visually identical: a top chord, a bottom chord and a triangulated web (figs. 1 and 4). The signed-difference plot in fig. 5 shows a slight redistribution along the boundary of each member, with a mean absolute density difference of 0.0024, an RMS difference of 0.0051 and a maximum of 0.0623. The fine-mesh compliance of the MTOP design exceeds the classical one by only 0.0601 %, which is well within the modeling error of the SIMP interpolation itself. This confirms that, for sensitivity filtering with $r_{\min} = 24$ density cells, the coarse 120×40 analysis mesh contains enough kinematic resolution to resolve the dominant load paths.

8.3 Origin of the speedup

The wall-clock time per iteration is dominated by the equilibrium solve. With sparse Cholesky factorization in two dimensions, the cost of solving $\mathbf{Ku} = \mathbf{f}$ scales approximately as $\mathcal{O}(N_{\text{dof}}^{3/2})$, so the ratio of analysis-mesh sizes is expected to translate into a speedup of order $(N_{\text{dof}}^{\text{cl}}/N_{\text{dof}}^{\text{mtop}})^{3/2} \approx 24.3^{3/2} \approx 120$. The measured speedup of 18.0 is markedly smaller because the MTOP iteration also pays for operations that scale with the density resolution rather than with the analysis mesh: the cone-kernel sensitivity filter on 600×200 values, the assembly of the multiresolution stiffness with 25 templates per analysis element, and the per-density-cell strain-energy evaluation (9). These per-cell operations are negligible in the classical run because they are fused with the FE assembly, which scales with the analysis mesh; in the MTOP run they constitute a non-trivial fraction of the per-iteration cost. The same trade-off is reported in [2] and is the main reason why the gain from MTOP is expected to grow further when the formulation is extended to three dimensions, where the FE solve cost grows faster than $\mathcal{O}(N_{\text{dof}}^{3/2})$ while the per-cell density operations remain quasi-linear.

8.4 Native versus fine-mesh compliance

The MTOP coarse-mesh compliance (219.352) is approximately 2.4 % below the fine-mesh compliance of the same design (224.731). This gap is not an error of the optimization but a consequence of the midpoint integration in eq. (7): with $N_c = 25$ uniformly placed integration points, the residual quadrature error of the bilinear element stiffness scales as $\mathcal{O}(h^2)$ with $h = 1/5$, which is consistent with the observed offset. The midpoint scheme is the natural choice for MTOP because it places one integration point per density cell, but it is less accurate than the analytical Q4 stiffness used by the classical formulation. The fine-mesh re-analysis is therefore the correct quantity for cross-comparison of the two designs. For the density-filtered runs the same offset is much smaller (2.3 %, from 235.621 to 241.028); the filter pre-smooths the densities before the SIMP analysis and the residual quadrature error is reduced.

8.5 Density filtering with vanilla optimality criteria

For density filtering both the classical and the MTOP runs hit the iteration cap of 300 without satisfying the design-change tolerance $\varepsilon = 0.01$: the maximum absolute change oscillates between 0.04 and 0.07 throughout the second half of the run, while the compliance is already stable to within 0.1 percent of its final value. The cause is the well-known incompatibility between the multiplicative OC update and density filtering on a fine mesh: the filter spreads each design-variable change over a neighbourhood of cells, but the OC update rescales every variable individually with no built-in damping, so high-frequency components of the design alternate sign from one iteration to the next. With sensitivity filtering the trick of dividing by $\max(\gamma, x_i)$ in eq. (10) damps this oscillation; with density filtering no such mechanism is available. The compliance plateau is therefore meaningful even though the change tolerance is not formally satisfied, and the move to MMA in step 4 is motivated precisely by this lack of robust convergence.

8.6 Density filtering versus sensitivity filtering as a comparison setting

Density filtering reduces the gap between the classical and the MTOP layouts even further than sensitivity filtering does. With sensitivity filtering the MTOP fine-mesh compliance lies 0.06 % above the classical one and the maximum density difference is 0.062; with density filtering the gap drops to 0.02 % and the maximum density difference to 0.016. Two effects reinforce each other. First, the density filter explicitly smooths the design variables before they enter the SIMP analysis, which makes the choice of integration scheme inside an analysis element (analytical for the classical formulation, 5×5 midpoint for MTOP) less consequential. Second, the smoother density distribution reduces the residual quadrature error of the multiresolution stiffness matrix (7), since this error scales with the density variation across density cells. The wall-clock speedup, on the other hand, drops from 18.0 to 6.32 because both density-filtered runs need a chain-rule convolution per OC step and an additional convolution inside the bisection (to evaluate $\rho^{\text{new}} = \mathcal{F}(x^{\text{new}})$ for the volume check); these per-density-cell operations scale with the density resolution and add a fixed cost that is the same for the classical and the MTOP runs and therefore reduces their ratio.

8.7 MMA versus optimality criteria

The MMA experiments of step 4 share the same MTOP analysis stack as the OC runs but use the method of moving asymptotes for the design update. Three observations are worth highlighting.

First, MMA does not by itself fix the change-tolerance issue of density filtering on this problem. Both the OC and the MMA density-filtered runs reach the iteration cap of 300 without satisfying $\max|\Delta x| < \varepsilon$, with very similar residual oscillation (about 0.05 for MMA vs. 0.04–0.07

for OC). The compliance plateaus to a well-defined value in both cases, and MMA produces a marginally better design ($-0.31\,\%$ on the fine mesh), but neither optimizer dampens the high-frequency components of the update enough to drive the change below 0.01.

Second, MMA does converge for sensitivity filtering, where OC also converged: the change drops below 0.01 after 188 iterations and the compliance is essentially identical to the OC value to within $0.8\,\%$ on the fine mesh. The slight increase comes from the fact that the sensitivity filter modifies $\partial C/\partial x_i$ heuristically rather than via a true chain rule, so the gradient passed to MMA is mildly inconsistent with the actual objective and the asymptote-based subproblem cannot exploit it as well as OC's monotonic update does. The number of iterations is therefore higher than OC's 94, but both reach a similar layout.

Third, the genuine advantage of MMA appears in the Heaviside experiments of Section 7.6: the projection introduces a sharply non-monotonic relationship between x and ρ, with derivatives that vanish away from $\widetilde{\rho} = \eta$ and peak strongly near it; OC's heuristic update would oscillate violently in this regime, whereas MMA's adaptive asymptotes track the projected response and allow the continuation strategy to succeed. The fine-mesh compliance drops from 240.253 (MMA density filter) to 199.472 for the symmetric $\eta = 0.5$ projection, a reduction of $17\,\%$, because the SIMP penalization is much more effective on a near-binary density field than on a gray density field of the same volume. The per-iteration cost of MMA is slightly higher than that of OC because of the interior-point subproblem, but this overhead is small compared with the FE solve and the per-density-cell operations.

8.8 Effect of the Heaviside threshold η

The Heaviside projection sharpens the boundary between solid and void: at $\beta = \beta_{\max} = 16$ the projected density ρ_i takes values close to 0 or 1 everywhere except in a narrow transition layer of width $O(\beta^{-1})$ around the threshold $\widetilde{\rho}_i = \eta$. The threshold controls the asymmetry of the projection. For $\eta < 0.5$ a smaller filtered density is sufficient to project a cell to solid, so under the volume constraint the optimizer produces a more conservative (lower) filtered density and the design enforces a minimum solid feature size of the order of the filter radius. For $\eta > 0.5$ the projection is biased towards void: the optimizer compensates by producing a richer filtered density, which enforces a minimum void feature size instead. For $\eta = 0.5$ both biases are absent and the projection is symmetric.

For the present problem the three Heaviside fine-mesh compliances differ by less than $0.7\,\%$: 200.129 for $\eta = 0.3$, 199.472 for $\eta = 0.5$ and 200.699 for $\eta = 0.7$. The symmetric threshold attains the lowest value, as expected for a problem in which neither solid nor void features are intrinsically more constrained than the other, but the difference is small enough that all three projections produce essentially the same load path. The visible distinctions between the three designs in fig. 15 are minor: the $\eta = 0.3$ run reaches a clean black-and-white state with no spurious gray spots, whereas at $\eta = 0.5$ and $\eta = 0.7$ a few small interior gray pockets remain at $\beta_{\max}$, which slightly increase the SIMP-penalised compliance even when their volume contribution is negligible. The continuation schedule on β keeps each subproblem well-posed: every doubling of β produces a small upward jump in compliance (the new projection is sharper than the design was optimized for) followed by a re-optimization plateau, which is visible in fig. 16.

9 Use of generative AI

GenAI was used during this work in three roles. First, it was used to set up the repository layout, the build of `run_assignment.m` and the LaTeX skeleton of this report. Second, it was used to translate the lecture's Chapter 9 examples `ex1a.m` (sensitivity and density filtering with OC) and `ex4.m` (MMA with sensitivity filter, density filter and Heaviside projection) into the production runners of steps 1–4; each resulting runner was checked against the lecture script line by line. Third, it was used to draft the multiresolution stiffness assembly, its compliance and Heaviside chain-rule sensitivities, and the verification routine of Section 5; the resulting expressions (7)–(9) and (12)–(13) were compared with the published derivations of [2] and [3], and the implementation was validated against finite-difference reference values for the per-density-cell SIMP sensitivities and for the full filter–projection chain to a relative tolerance below 10^{-5}. All numerical results, figures and physical interpretations were generated and inspected by the author and remain the author's responsibility.

10 Conclusion

Step 1 establishes the classical OC reference solution for the 600×200 MBB beam with sensitivity filtering. Step 2 shows that the multiresolution scheme of Nguyen et al. with a 120×40 analysis mesh and 5×5 density cells per analysis element reproduces the classical layout to within 0.06 % of fine-mesh compliance and reduces the optimization time from 241.99 s to 13.43 s, a speedup factor of 18.0. Step 3 repeats the comparison with density filtering: the MTOP design now matches the classical density-filtered design to within 0.02 % of fine-mesh compliance and the wall-clock speedup is 6.32. The reduction of both the design gap and the speedup is consistent: density filtering pre-smooths the design variables, which makes the multiresolution integration error smaller but also adds chain-rule convolutions whose cost scales with the density resolution and is the same for both formulations. Vanilla OC fails to satisfy the design-change tolerance for either density-filtered run.

Step 4 replaces OC with MMA inside the multiresolution scheme and investigates three filter choices. With sensitivity filtering MMA satisfies the change tolerance in 188 iterations and reaches a fine-mesh compliance of 226.452, very close to the OC value of 224.731 but 0.77 % higher because the sensitivity-filter heuristic is not an exact chain rule for MMA. With density filtering MMA still hits the iteration cap of 300 with a residual change of about 0.05; switching the optimizer alone does not eliminate the convergence pathology of step 3. The genuine advantage of MMA appears once Heaviside projection is added: the three runs at $\eta \in \{0.3, 0.5, 0.7\}$ converge cleanly under continuation on β and produce essentially black-and-white designs at $\beta = \beta_{\max} = 16$. The symmetric threshold $\eta = 0.5$ achieves the lowest fine-mesh compliance of 199.472, a reduction of 17 % compared with the gray density-filtered design at the same volume fraction, and the asymmetric thresholds shift the layout towards thicker or slimmer members in the way predicted by the minimum-feature-size interpretation of the projection.

References

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